

Contact

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(LinkedIn)

Top Skills

Machine Learning
Amazon Web Services (AWS)
RAG

Languages

English
Chinese (Native or Bilingual)

Certifications

Algorithms: Design and Analysis,
Part 2
AI for Medical Diagnosis
Introduction to Angular2
Blockchain Developer - NanoDegree
Generative AI with Large Language
Models

Publications

Combined small molecule inhibition
accelerates developmental timing
and converts human pluripotent stem
cells into nociceptors
Specification of functional cranial
placode derivatives from human
pluripotent stem cells

Lei Niu

Full-Stack AI Solutions Tech Lead
Scarsdale, New York, United States

Summary

Tech Lead | AI Believer | PhD

Experience

Vanilla

Staff Software Engineer

January 2025 - Present (7 months)

- Established an end-to-end local development workflow and successfully onboarded the team within the first two weeks. This initiative improved development efficiency by over 20%, enabling seamless local testing and reducing dependence on edge environments.
- Designed and deployed a robust LLM agent pipeline in the second month, significantly strengthening the team's AI infrastructure and laying the groundwork for scalable innovation.
- Independently developed and launched an AI-powered feature for auto-generating provision notes, achieving production deployment within four months. This solution automates 32–48% of the manual abstraction process, resulting in substantial time savings and streamlining workflows across the abstraction lifecycle.
- Led a one-day, two-person hackathon project in the fifth month to build an internal tool that auto-generates Jira tickets, dramatically reducing the customer support team's manual workload. Oversaw the deployment process to ensure successful production rollout.
- Reduced deployment build time by 90% and total deployment duration by 60%

Sotto

Principal Software Engineer

October 2023 - January 2025 (1 year 4 months)

New York, New York, United States

(formerly BigCo)

- Architected and implemented advanced AI solutions leveraging Large Language Models (LLM) and Retrieval-Augmented Generation (RAG)

on AWS, driving transformation in customer engagement, operational optimization, and delivery of data-centric insights.

- Developed and maintained a suite of SaaS web services, including:

-- Datastore

-- Websocket

-- Comprehensive internal and external APIs (including Webhook infrastructure)

-- Feature Flag Infrastructure

-- Segmentation

-- Cloud messaging (Firebase Cloud Messaging - FCM)

-- Web scraper

-- Store-finder

-- vCard generator

-- Quiet hour checker

-- Scheduler

- Designed and executed seamless integrations with third-party services to expand platform capabilities and support diverse client requirements.

Haven Technologies

7 years 11 months

Solutions Architect

January 2023 - August 2023 (8 months)

Designs and implements SaaS solutions (including APIs, configurable micro services, data standard, et.al.), considering clients' requirements and goals.

Collaborate with stakeholders, evaluate technologies, and provide guidance for the successful deployment.

Contributed to the [ACORD Next-Generation Digital Standards](<https://www.acord.org/standards-architecture/acord-data-standards/next-generation-digital-standards>)

Chapter Lead (Engineer Manager)

May 2020 - January 2023 (2 years 9 months)

Managed eight engineers to support their performance improvement and help them achieve personal growth. As a result, four members were promoted.

Tech Lead

August 2019 - January 2023 (3 years 6 months)

Led 20+ Engineers in the Risk Solutions (Underwriting Solutions and Actuarial Solutions) team, and collaborated with Product Owners to deliver high-quality software solutions and foster a friendly and collaborative working environment. Major products include the Digital Underwriting Platform (Algo + Manual), Rules Engine, Document Digitization Service (OCR), ADAPT Actuarial Tool, and more.

Team Lead

June 2019 - January 2023 (3 years 8 months)

As the Team Lead of Team Panda, collaborated on projects with MMI in Asia. The team, consisting of five part-time developers, successfully achieved 5% of the company's annual goals of 2019 within seven months.

Interim Product Owner

May 2021 - December 2021 (8 months)

Greater New York City Area

Interim Product Owner, PORUS (Productization Of Resourceful Underwriting Solutions).

Application Developer

October 2015 - August 2019 (3 years 11 months)

Greater New York City Area

Software Engineer, Underwriting Team

Montefiore Information Technology

Software Engineer

September 2014 - October 2015 (1 year 2 months)

Design and develop the backend and UI of a powerful big data platform, which provides the convenient access to billions of data.

Albert Einstein College of Medicine

Senior Research Fellow

December 2011 - August 2014 (2 years 9 months)

Implement generating System, record the big data, and analyze the data with multi-statistic tools.

Technologies: Matlab, big data, statistic

Memorial Sloan Kettering Cancer Center

Research Fellow

August 2009 - November 2011 (2 years 4 months)

Education

University of Science and Technology of China

Doctor of Philosophy (Ph.D.), Biophysics · (2003 - 2008)

Georgia Institute of Technology

Other, Computer Science · (2014 - 2014)

University of Science and Technology of China

Bachelor of Engineering (BEng), Computer Science and
Technology · (2000 - 2003)

University of Science and Technology of China

Bachelor of Science (BS), Biological Science · (1998 - 2003)